

# MARILENE KOUGOUM MOKO MANI

Looking for a 6 months internship in quantitative research, trading or structuring starting April 2026

Visit my personal website for portfolio and projects

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🐙 [my-github](https://github.com/my-github)

## Education

### Master M2MO (Formerly DEA Laure ELIE)

Master degree in Stochastic modelling, Quantitative finance and Data science

September 2025 – September 2026

Paris, France

### ENSAI Rennes

Engineering degree in Risk management and Data science

September 2023 – September 2025

Rennes, France

## Technical Skills

- **Methods:** Applied mathematics, stochastic processes, derivative pricing hedging, stochastic model calibration, numerical methods (Monte Carlo, PDEs), numerical optimization, risk & asset management, algorithmic trading, execution strategies, market data analysis, machine learning, deep learning, big data & NLP .
  - **Programming:** Python(pandas, polars, numpy, scikit-learn, TensorFlow), C++, R, Matlab, PySpark, SQL, VBA.
- Soft skills:** Delivery-driven mindset, proactive problem-solving approach, attention to detail, pressure-resilient.

## Experience

### Quantitative analyst | Société Générale

Model Risk Department

April 2025 – September 2025

La Défense, France

- \* Led Société Générale's official response to an IGAD post-audit recommendation on the robustness of the climate stress-testing model (non-retail), delivered within a 3-month regulatory deadline.
  - Developed distributional stress and sensitivity analyses by simulating input perturbations and quantifying their propagation to output risk metrics.
  - Built an automated robustness-testing framework for climate stress-test validation and drafted the associated regulatory response deliverable.
- \* Designed and implemented an alternative IFRS9 risk-parameter calibration methodology; benchmarked it against the production model, analyzed economic consistency, stability, and assessed impacts on provisions and risk profile.

### Statistical research internship | DARES - UNEDIC

Employment and Labour Market Department

June 2024 – September 2024

Paris 15, France

- \* Conducted a causal analysis of the impact of natural disasters on short-time work uptake, in collaboration with Dr. Oana Calavrezo.
  - Applied statistical and econometric methods (2SLS) and implemented propensity-score matching to identify causal effects and reduce selection bias, proposed and tested multiple instruments.

## Projects

### Yield curve modelling ([View project on website](#)) | Python

March 2025

- \* Reconstructed zero-coupon curves via bootstrapping and interpolation; priced vanilla caplets using Black model and calibrated implied volatilities.
- \* Calibrated Hull & White model (HJM framework) and simulated Bermudan barrier interest rate options via Monte Carlo.

### Stock price forecasting with Recurrent Neural Networks ([Implementation on GitHub](#)) | Python Feb. 2025

- \* Implemented LSTM networks for non-linear dynamics and long-range temporal dependencies; benchmarked against Gradient Boosting, Ridge regression and Facebook Prophet models.
- \* Deployed best-performing model as an interactive Streamlit app for real-time visualization ([View the app](#)).

### Asset Pricing & Management ([Implementation on GitHub](#)) | Python, Dash

February 2025

- \* Developed option pricing frameworks for European, American, and Asian options with Black-Scholes; computed Greeks for risk management ([View on website](#)) .
- \* Designed Markowitz portfolio optimization strategies and interactive Dash dashboards for real-time analytics ([view](#)).

## Languages & Certifications

**French:** Native **English:** Fluent (TOEIC certified)

**Certifications:** AMF (Autorité des marchés financiers), TOEIC (910/990 : B2+ )